Cell Tuition
1-1 PRIVATE TUTORS

NUMBER

Ratios

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KEY WORDS ratio *(ōwehenga)* · part · whole · simplest form *(āhua māmā rawa)* · equivalent ratio · share

A Writing ratios ●○○

⚠ **Watch out:** The order matters. The ratio of red to blue is not the same as the ratio of blue to red.

1. A bag holds 3 red counters and 5 blue counters. Write each ratio.

a) red to blue _____ **b)** blue to red _____ **c)** red to all the counters _____

2. A fruit bowl holds 5 apples, 4 pears and 2 oranges. Write each ratio.

a) apples to pears _____ **b)** oranges to apples _____ **c)** pears to all the fruit _____

B Simplifying ratios ●○○

Remember: Divide both parts by the same number, and keep going until no number divides into both: $12 : 8 = 3 : 2$.

3. Write each ratio in its simplest form.

a) 4 : 6 = _____ **b)** 10 : 5 = _____ **c)** 9 : 12 = _____ **d)** 8 : 20 = _____

e) 15 : 25 = _____ **f)** 14 : 21 = _____ **g)** 18 : 12 = _____ **h)** 30 : 45 = _____

i) 24 : 16 = _____ **j)** 50 : 20 = _____ **k)** 36 : 48 = _____ **l)** 6 : 9 : 12 = _____

C Equivalent ratios ●●○

4. Find the missing number.

a) $1 : 2 = 4 : \square$ **b)** $2 : 3 = \square : 9$ **c)** $3 : 5 = 9 : \square$ **d)** $4 : 1 = \square : 5$

e) $2 : 7 = 10 : \square$ **f)** $\square : 3 = 25 : 15$ **g)** $3 : \square = 21 : 28$ **h)** $6 : 5 = 30 : \square$

D Sharing in a ratio ●●○

⚠ **Watch out:** Add the parts of the ratio first. To share in the ratio 2 : 3 there are 5 parts, so divide the amount by 5, not by 2 or by 3.

5. Share each amount.

a) Share \$20 in the ratio 1 : 3.

b) Share \$30 in the ratio 2 : 3.

c) Share 42 kg in the ratio 1 : 5.

d) Share 60 m in the ratio 3 : 7.

e) Share \$45 in the ratio 4 : 5.

f) Share 96 cm in the ratio 5 : 3.

g) Share \$250 in the ratio 3 : 2.

h) Share 1.2 litres in the ratio 1 : 2.

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E Part-to-whole ratios ●●●

> **Remember:** A ratio can compare a part with another part, or a part with the whole. If boys to girls is 2 : 3, there are 5 parts altogether, so boys to all students is 2 : 5.

6. Answer each question.

a) In a class, the ratio of boys to girls is 2 : 3. Write the ratio of boys to all the students.

b) In a box, the ratio of red beads to all the beads is 3 : 10. Write the ratio of red beads to the other beads.

c) Juice is mixed from concentrate and water in the ratio 1 : 4. Write the ratio of concentrate to all the juice.

d) In a school choir, the ratio of boys to all the singers is 2 : 5. There are 30 singers. How many of them are boys?

e) In a car park, the ratio of white cars to other cars is 3 : 4. There are 28 cars altogether. How many of them are white?

F Ratios, fractions and percentages ●●●

> ⚠ **Watch out:** A ratio of 1 : 3 does not mean one third. There are 4 parts altogether, so the first part is $\frac{1}{4}$, which is 25%.

7. Answer each question.

a) Paint is mixed from blue and white in the ratio 1 : 3. What fraction of the paint is blue?

b) The ratio of adults to children at a party is 2 : 3. What fraction of the people are children?

c) The ratio of wins to losses for a team is 3 : 7. What percentage of its games did the team win?

d) $\frac{3}{5}$ of the students in a class have a pet. Write the ratio of students with a pet to students without a pet.

e) Sand and cement are mixed in the ratio 4 : 1. What percentage of the mix is cement?

★ Extension

8. The ratio of cats to dogs at a vet is 3 : 4. There are 12 cats. How many dogs are there?

9. Share \$40 in the ratio 2 : 3 : 5.

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Cell Tuition
1-1 PRIVATE TUTORS

NUMBER

Ratios

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS ratio *(ōwehenga)* · part · whole · simplest form *(āhua māmā rawa)* · equivalent ratio · share

A Writing ratios ●○○

⚠ **Watch out:** The order matters. The ratio of red to blue is not the same as the ratio of blue to red.

1. A bag holds 4 red counters and 7 blue counters. Write each ratio.

a) red to blue _____ **b)** blue to red _____ **c)** red to all the counters _____

2. A fruit bowl holds 3 apples, 5 pears and 4 oranges. Write each ratio.

a) apples to pears _____ **b)** oranges to apples _____ **c)** pears to all the fruit _____

B Simplifying ratios ●○○

Remember: Divide both parts by the same number, and keep going until no number divides into both: $12 : 8 = 3 : 2$.

3. Write each ratio in its simplest form.

a) 6 : 8 = _____ **b)** 12 : 4 = _____ **c)** 6 : 15 = _____ **d)** 8 : 12 = _____

e) 20 : 25 = _____ **f)** 21 : 28 = _____ **g)** 20 : 12 = _____ **h)** 30 : 50 = _____

i) 32 : 24 = _____ **j)** 70 : 20 = _____ **k)** 27 : 36 = _____ **l)** 4 : 8 : 10 = _____

C Equivalent ratios ●●○

4. Find the missing number.

a) $1 : 3 = 4 : \square$ **b)** $2 : 5 = \square : 15$ **c)** $3 : 4 = 9 : \square$ **d)** $5 : 1 = \square : 4$

e) $3 : 7 = 15 : \square$ **f)** $\square : 3 = 24 : 18$ **g)** $2 : \square = 14 : 35$ **h)** $7 : 4 = 35 : \square$

D Sharing in a ratio ●●○

⚠ **Watch out:** Add the parts of the ratio first. To share in the ratio 2 : 3 there are 5 parts, so divide the amount by 5, not by 2 or by 3.

5. Share each amount.

a) Share \$24 in the ratio 1 : 3.

b) Share \$35 in the ratio 2 : 3.

c) Share 48 kg in the ratio 1 : 5.

d) Share 80 m in the ratio 3 : 7.

e) Share \$63 in the ratio 4 : 5.

f) Share 72 cm in the ratio 5 : 3.

g) Share \$350 in the ratio 3 : 2.

h) Share 1.5 litres in the ratio 1 : 2.

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E Part-to-whole ratios ●●●

> **Remember:** A ratio can compare a part with another part, or a part with the whole. If boys to girls is 2 : 3, there are 5 parts altogether, so boys to all students is 2 : 5.

6. Answer each question.

a) In a class, the ratio of boys to girls is 3 : 4. Write the ratio of boys to all the students.

b) In a box, the ratio of red beads to all the beads is 2 : 9. Write the ratio of red beads to the other beads.

c) Juice is mixed from concentrate and water in the ratio 1 : 3. Write the ratio of concentrate to all the juice.

d) In a school choir, the ratio of boys to all the singers is 3 : 5. There are 40 singers. How many of them are boys?

e) In a car park, the ratio of white cars to other cars is 2 : 5. There are 35 cars altogether. How many of them are white?

F Ratios, fractions and percentages ●●●

> ⚠ **Watch out:** A ratio of 1 : 3 does not mean one third. There are 4 parts altogether, so the first part is $\frac{1}{4}$, which is 25%.

7. Answer each question.

a) Paint is mixed from blue and white in the ratio 1 : 4. What fraction of the paint is blue?

b) The ratio of adults to children at a party is 3 : 2. What fraction of the people are children?

c) The ratio of wins to losses for a team is 7 : 3. What percentage of its games did the team win?

d) $\frac{2}{5}$ of the students in a class have a pet. Write the ratio of students with a pet to students without a pet.

e) Sand and cement are mixed in the ratio 9 : 1. What percentage of the mix is cement?

★ Extension

8. The ratio of cats to dogs at a vet is 2 : 5. There are 10 cats. How many dogs are there?

9. Share \$30 in the ratio 1 : 2 : 3.

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